9.2 Resistance

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.2 Resistance
Difficulty	Medium

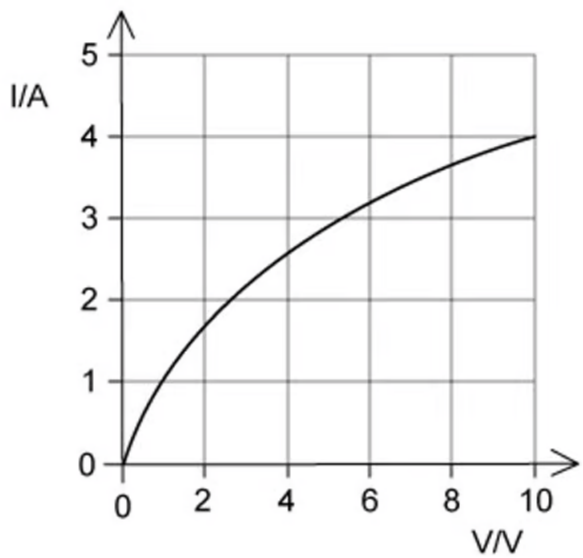
Time allowed: 20

Score: /10

Percentage: /100

Question 1

The graph shows how current *I* varies with voltage *V* for a filament lamp.



Since the graph is not a straight line, the resistance of the lamp varies with *V*.

Which row gives the correct resistance at the stated value of *V*?

	V / V	**R / Ω**
A	2.0	1.5
B	4.0	3.2
C	6.0	1.9
D	8.0	0.9

[1 mark]

Question 2

A charge of 8.0 C passes through a resistor of resistance 30 Ω at a constant rate in a time of 20 s.

What is the potential difference across the resistor?

A. 0.40 V

B. 5.3 V

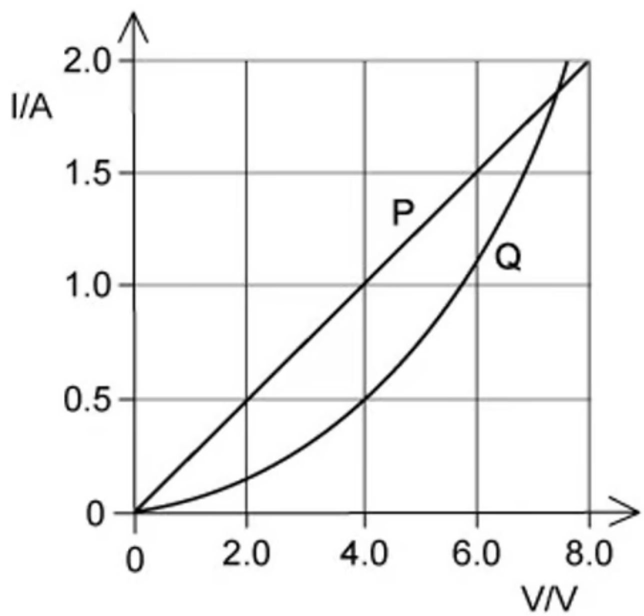
C. 12 V

D. 75 V

[1 mark]

Question 3

The I-V characteristics of two electrical components P and Q are shown below.



Which statement is correct ?

A. P is a resistor and Q is a filament lamp

B. the resistance of Q increases as the current in it increases

C. for a current of 1.9 A, the resistance of Q is approximately half that of P

D. for a current of 0.5 A, the power dissipated in Q is double that in P

[1 mark]

Question 4

The resistance R of a resistor is to be determined. The current I in the resistor and the potential difference V across it are measured.

The results, with their uncertainties, are

$I = (2.0 \pm 0.2)\,\text{A}$ $\quad\quad V = (15.0 \pm 0.5)\,\text{V}$

What is the uncertainty in this value for R?

A. $\pm 0.3\,\Omega$

B. $\pm 0.5\,\Omega$

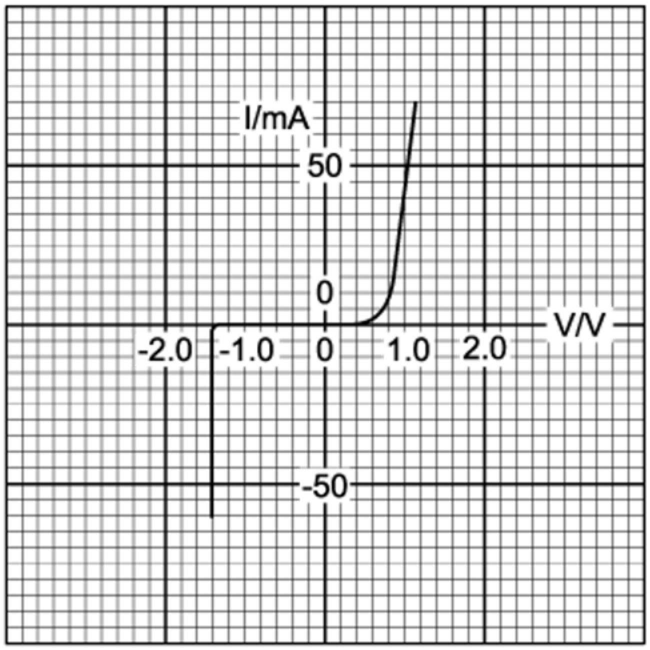
C. $\pm 0.7\,\Omega$

D. $\pm 1\,\Omega$

[1 mark]

Question 5

The variation with potential difference V of the current I in a semiconductor diode is shown below.



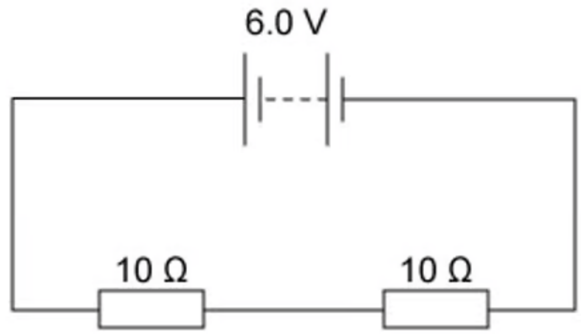
What is the resistance of the diode for applied potential differences of +1.0 V and –1.0 V?

	Resistance	
	at +1.0 V	**at –1.0 V**
A	20 Ω	infinite
B	20 Ω	zero
C	0.05 Ω	infinite
D	0.05 Ω	zero

[1 mark]

Question 6

A battery of negligible internal resistance is connected to two 10 Ω resistors in series.



What charge flows through each of the 10 Ω resistors in 1 minute?

A. 0.30 C

B. 0.60 C

C. 3.0 C

D. 18 C

[1 mark]

Question 7

Two heating coils X and Y, of resistance R_X and R_Y respectively, deliver the same power when 12 V is applied across X and 6 V is applied across Y.

What is the ratio $\frac{R_X}{R_Y}$?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 2

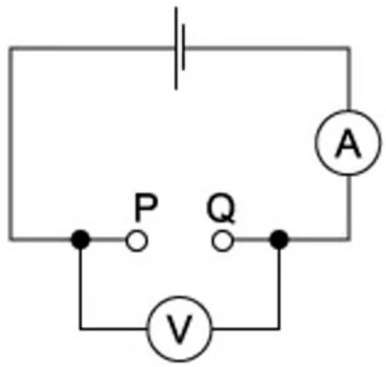
D. 4

[1 mark]

Question 8

A student found two unmarked resistors. To determine the resistance of the resistors, the circuit below was set up. The resistors were connected in turn between P and Q, noting the current readings.

The voltage readings were noted without the resistors and with each resistor in turn.



The results were entered into a spreadsheet, as shown.

1.5	1.3	28	46
1.5	1.4	14	100

The student forgot to enter the column headings.

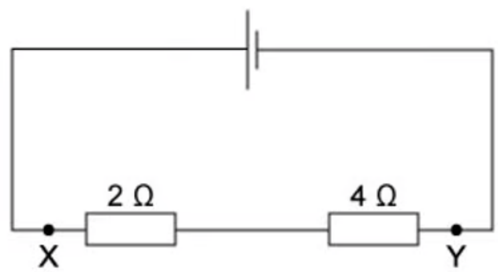
Which order of the headings would be correct?

A	e.m.f / V	V / V	R / Ω	I / mA
B	V / V	e.m.f / V	R / Ω	I / mA
C	V / V	e.m.f / V	I / mA	R / Ω
D	e.m.f / V	V / V	I / mA	R / Ω

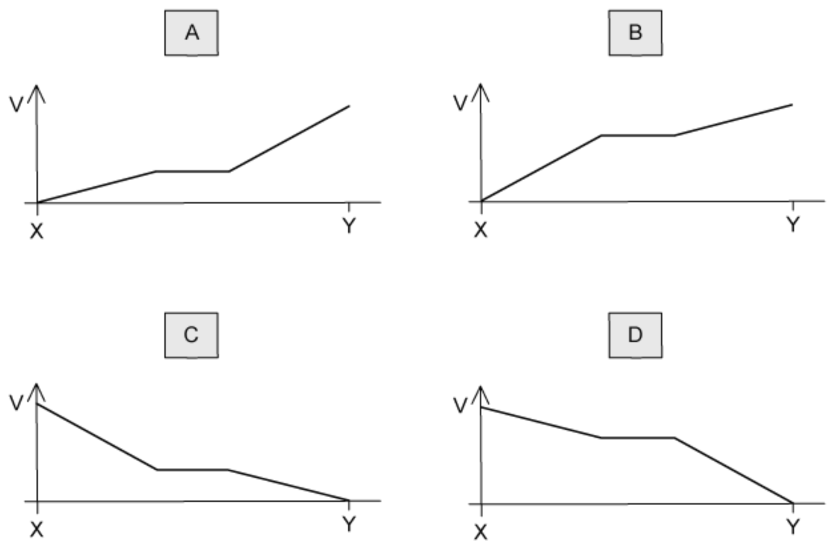
[1 mark]

Question 9

A 2 Ω resistor and a 4 Ω resistor are connected to a cell.



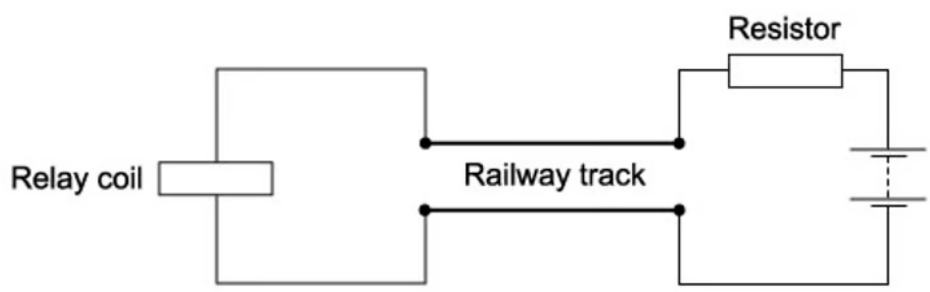
Which graph shows how the potential V varies with distance between X and Y ?



[1 mark]

Question 10

The diagram shows a length of track from a model railway connected to a battery, a resistor and a relay coil.



With no train present, there is a current in the relay coil which operates a switch to turn on a light.

When a train occupies the section of track, most of the current flows through the wheels and axles of the train in preference to the relay coil. The switch in the relay turns off the light.

Why is a resistor placed between the battery and the track?

A. to limit the heating of the wheels of the train

B. to limit the energy lost in the relay coil when a train is present

C. to prevent a short circuit of the battery when a train is present

D. to protect the relay when a train is present

[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

- Ohm's Law is defined by the equation $V = IR$
- A filament lamp does not follow Ohm's Law since the variation of I with V is not linear, and a straight line is not obtained
- Resistance is equal to $R = \frac{V}{I}$
- Consider option **C**:
  - At V = 6.0 V, current I = 3.2 A
  - Resistance $R = \frac{V}{I} = \frac{6.0}{3.2} = 1.875\,\Omega$
- Therefore, the resistance at 6.0 V is 1.9 Ω; this matches option **C** correctly

A is incorrect because At V = 2.0 V, current I = 1.8 A, Resistance $R = \frac{2.0}{1.8} = 1.11\,\Omega$

This value ≠ 1.5 Ω

B is incorrect because At V = 4.0 V, current I = 2.5 A, Resistance $R = \frac{4.0}{2.5} = 1.6\,\Omega$

This value ≠ 3.2 Ω

D is incorrect because At V = 8.0 V, current I = 3.6 A, Resistance $R = \frac{8.0}{3.6} = 2.2\,\Omega$

This value ≠ 0.9 Ω

Exam Tip: The values of current *I* read from the graph may not be exact for all the points, but they are close to the correct value. You do not need to be extra precise in these types of question (but you do in the written answer questions!)

2

The correct answer is **C** because:

- Potential difference across a resistor is equal to $V = IR$
- Current is equal to $I = \frac{Q}{t}$
- Combining these equations gives $V = \frac{QR}{t}$
  - So, the potential difference is $V = \frac{8.0 \times 3.0}{20} = 12\ V$

3

The correct answer is **D** because:

- Graph P shows a component which obeys Ohm's Law
- Graph Q shows a component whose resistance decreases with *V*, unlike the graph for a filament lamp
- Power dissipated is equal to $P = VI$
- At current I = 0.5 A
  - Graph P: *V* = 2.0 V
  - Graph Q: *V* = 4.0 V
- So, for graph Q, *V* is twice that of graph P ; therefore the power dissipated in Q is double that in P
  - Since $(2V)I = 2P$

A is incorrect because the I-*V* characteristics of a filament lamp would should a decreasing gradient, unlike the graph Q, which has an increasing gradient. This is because as *V* increases, the temperature of the lamp increases, causing the resistance to increase

B is incorrect because as *V* increases, the current I increases. From Ohm's Law, $I = \frac{V}{R}$, so the resistance *R* should be getting smaller

C is incorrect because at current I = 1.9 A, the 2 graphs intercept, so, they have the same value of *V* at that point, and hence the same resistance

4

The correct answer is **D** because:

- The value of *R* is $R = \frac{V}{I} = \frac{15.0}{2.0} = 7.5\Omega$

- From the equation , the formula to work out the uncertainty in resistance is
  - $\left(\frac{\Delta R}{R}\right)=\left(\frac{\Delta V}{V}\right)+\left(\frac{\Delta I}{I}\right)$
- To work out uncertainty in data which is multiplied or divided we add the uncertainties
  - $\left(\frac{\Delta R}{R}\right)=\left(\frac{0.5}{15.0}\right)+\left(\frac{0.2}{2.0}\right)=\frac{2}{15}$
- Then, to find the uncertainty in *R* we multiply by the calculated value of *R*
  - $\Delta R=\frac{2}{15}\times 7.5=\pm 1\,\Omega$

5

The correct answer is **A** because:

- A semiconductor diode is a device that has very low resistance in one direction and a very high resistance in the other direction
  - Think of it like a ball at the top of a hill: it can roll from top to bottom easily, but requires external energy to take it from the bottom to the top of the hill

- When there is no current, the resistance is infinite. Because of this, the electrical resistance of the diode is very high or infinite for negative voltages until about –1.5 V

- The electrical resistance of the diode decreases from 0.5 V onwards
- At 1.0 V, the current is equal to 50 mA or 0.05 A
  - So, the resistance at this point is $R=\frac{V}{I}=\frac{1.0}{0.05}=20\,\Omega$

6

The correct answer is **D** because:

- Potential difference across a resistor is equal to $V=IR$
- Current is equal to $I=\frac{Q}{t}$
- Combining these equations gives $V=\frac{QR}{t}$
- Rearranging this for charge *Q* gives $Q=\frac{Vt}{R}$
  - The p.d. through each 10 Ω resistor is $V=3.0\ V$
  - The time is equal to $t=60\ s$

- So, $Q = \frac{Vt}{R} = \frac{3.0 \times 60}{10} = 18\ C$

7

The correct answer is **D** because:

- Electrical power is defined as $P = VI$
- Ohm's Law is given by $V = IR$
- Substituting I for $I = \frac{V}{R}$
  - $P = V\left(\frac{V}{R}\right) = \frac{V^2}{R}$
- Power delivered by X = Power delivered by Y
  - $P_X = P_Y$
  - $\frac{12^2}{R_X} = \frac{6^2}{R_Y}$
  - $\frac{R_X}{R_Y} = \frac{144}{36} = 4$

8

The correct answer is **D** because:

- The e. m .f. of the supply in the circuit is constant and independent of the other components in the circuit. Without the resistors in the circuit, the voltmeter will read the e .m. f. of the supply. Since the e .m .f is constant, the first column, which is the only column with constant values, must be the e. m .f.
- The second column, which contain values close to 1.5, must be the p .d. across the resistor
- The values in this column increases from 1.3 to 1.4 when the values of the last column increase from 46 to 100
- Ohm's Law states $V = IR$, so as p. d. increases, *R* increases, so the final column must be *R*
- Additionally, it can be seen that when the resistance *R* is greater, the current is smaller

A is incorrect because if you consider the voltage drop due to internal resistance, equal to Ir, there would be a more significant drop when the current is greater, but this wouldn't make sense if the p. d. increased from 1.3 to 1.4 V if the current increased from 46 – 100 mA

B & C are incorrect because the first column must be e. m .f. as this value does not change

Exam Tip: Note that usually, when filling columns, as in this example, the first column would be the values that remain constant (if any – here it is the e .m .f.) and the last column is a quantity that needs to be calculated (here, the resistance). The middle columns usually contain the measured quantities (here, current and p .d.).

9

The correct answer is **D** because:

- The cell has some e .m .f., say *E*. The negative terminal of the cell is usually taken to be at a potential of 0 V while the positive terminal is taken to be at a potential of *E*
- So, the positive terminal is at a higher potential than the negative terminal
- No component is connected between the positive terminal and point X, so they are at the same potential
- Therefore, point X is at potential E, while, point Y is at the same potential as the negative terminal of the cell, so, the potential at Y is zero
- Ohm's Law states $V = IR$
- Therefore, the p. d. across the 2Ω resistor must be less than the p .d. across the 4Ω resistor. That is, the change in potential across the 4Ω resistor is more significant than that across the 2Ω resistor
- In the graph, this is represented by the gradient. The gradient across the 4Ω resistor, which is just before point Y, is steeper than that across the 2Ω resistor which is just after point X

A & B are incorrect because the potential at point X is *E* and point Y is zero so the gradient should decrease

C is incorrect because Ohm's Law tells us that $V \alpha R$, so the gradient is steeper across the 4Ω resistor and shallower across the 2Ω resistor

10

The correct answer is **C** because:

- When the train is present, it is stated that most of the current flows through the wheels and axles of the train in preference to the relay coil
- So, there is little current in the relay coil; therefore, the energy lost in the relay coil when a train is present is already limited
- Ohm's Law states $V = IR$, so the link between current and resistance is $I = \frac{V}{R}$
- The resistor actually limits the current flowing in the circuit when the train is present

- Therefore, this prevents a short circuit of the battery when a train is present

A is incorrect because the heating of the wheels due to the current is small compared to the heating due to its motion

B & D are incorrect because the relay is already protected, so this isn't the reason for needing a resistor